

The Matrix-Square Case of the All-Level Line-Plus-Simplex Conjecture

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Source Problem

The source is Bluhm–Evert–Klep–Magron–Nechita, *Inclusion constants for free spectrahedra with applications to quantum incompatibility*, arXiv:2512.17706 [1]. The paper proves a level-two formula for the direct sum of a real line and a real k -simplex and then conjectures the same value without a level restriction:

We in fact conjecture that the above result holds even if one does not restrict to level two of $\mathcal{D}_{A(k)}$ and $\mathcal{D}_{A(k)}^\square$. That is, we conjecture that the optimum is still $\frac{2k}{k-1+\sqrt{1+k}}$ even if one searches over all of $\mathcal{D}_{A(k)}$ and $\mathcal{D}_{A(k)}^\square$.

Conjecture 5.2. *Let $\mathcal{D}_{A(k)}$ be as defined in Theorem 5.1. Then the maximum value achieved by*

$$\begin{aligned} \max \quad & \lambda_{\max} \sum (X_i \otimes Y_i) \\ \text{s.t.} \quad & X \in \mathcal{D}_A, \quad Y \in \mathcal{D}_A^\square. \end{aligned} \tag{35}$$

is equal to

$$\gamma(k) = \frac{2k}{k-1+\sqrt{1+k}}.$$

Furthermore, this maximum value is achieved when X is defined by

$$X_1 = \begin{pmatrix} 1 & 0 \\ 0 & -k \end{pmatrix}, \quad X_2 = \begin{pmatrix} -k & 0 \\ 0 & 1 \end{pmatrix}, \quad X_{k+1} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

and $X_j = I_2$ for all $j \neq 1, 2, k+1$ and when Y is defined by

$$Y_1 = \begin{pmatrix} \frac{1+\sqrt{1+k}}{1+k+2\sqrt{1+k}} & 0 \\ 0 & -\frac{1}{1+k+2\sqrt{1+k}} \end{pmatrix}, \quad Y_2 = \begin{pmatrix} -\frac{1}{1+k+2\sqrt{1+k}} & 0 \\ 0 & \frac{1+\sqrt{1+k}}{1+k+2\sqrt{1+k}} \end{pmatrix}, \quad Y_{k+1} = \begin{pmatrix} 0 & \frac{k}{k-1+\sqrt{1+k}} \\ \frac{k}{k-1+\sqrt{1+k}} & 0 \end{pmatrix},$$

and $Y_i = 0$ for all $i \neq 1, 2, k+1$.

This packet proves the $k = 1$ case of that all-level conjecture. Under the source definitions, $k = 1$ is the matrix square and its dual matrix diamond.

Result

Let

$$\mathcal{C}_2(n) = \{(X_1, X_2) \in M_n(\mathbb{C})_{\text{sa}}^2 : -I \leq X_i \leq I, \ i = 1, 2\}$$

be the two-variable matrix cube, and let

$$\mathcal{D}_2(n) = \{(Y_1, Y_2) \in M_n(\mathbb{C})_{\text{sa}}^2 : \epsilon_1 Y_1 + \epsilon_2 Y_2 \preceq I \text{ for every } \epsilon \in \{\pm 1\}^2\}$$

be the matrix diamond. These are exactly $\mathcal{D}_{A(1)}$ and $\mathcal{D}_{A(1)}^\square$ in Conjecture 5.2 of [1].

Theorem 1. *One has*

$$\sup_{X \in \mathcal{C}_2, Y \in \mathcal{D}_2} \lambda_{\max}(X_1 \otimes Y_1 + X_2 \otimes Y_2) = \sqrt{2}.$$

Consequently Conjecture 5.2 of [1] holds for $k = 1$, since

$$\gamma(1) = \frac{2}{\sqrt{2}} = \sqrt{2}.$$

Proof Intuition

The $k = 1$ dual constraints have a particularly simple form. The diamond condition is equivalent to saying that $A = Y_1 + Y_2$ and $B = Y_1 - Y_2$ are self-adjoint contractions. On the cube side, the combinations

$$P = \frac{X_1 + X_2}{2}, \quad Q = \frac{X_1 - X_2}{2}$$

satisfy $P^2 + Q^2 \leq I$. Thus the objective is a product of one row contraction and one column with norm at most $\sqrt{2}$. Equality is the standard CHSH/Pauli configuration.

Proof

Proof. Let $X = (X_1, X_2) \in \mathcal{C}_2(n)$ and $Y = (Y_1, Y_2) \in \mathcal{D}_2(m)$. Put

$$A = Y_1 + Y_2, \quad B = Y_1 - Y_2.$$

The four sign inequalities defining $\mathcal{D}_2$ are exactly

$$-I \leq A \leq I, \quad -I \leq B \leq I,$$

so A and B are self-adjoint contractions.

Next put

$$P = \frac{X_1 + X_2}{2}, \quad Q = \frac{X_1 - X_2}{2}.$$

Since X_1 and X_2 are self-adjoint contractions,

$$P^2 + Q^2 = \frac{X_1^2 + X_2^2}{2} \preceq I.$$

Moreover

$$X_1 \otimes Y_1 + X_2 \otimes Y_2 = P \otimes A + Q \otimes B.$$

Writing this as a row-column product,

$$P \otimes A + Q \otimes B = \begin{bmatrix} P \otimes I_m & Q \otimes I_m \end{bmatrix} \begin{bmatrix} I_n \otimes A \\ I_n \otimes B \end{bmatrix},$$

we obtain

$$\begin{aligned} \|P \otimes A + Q \otimes B\| &\leq \left\| \begin{bmatrix} P \otimes I_m & Q \otimes I_m \end{bmatrix} \right\| \left\| \begin{bmatrix} I_n \otimes A \\ I_n \otimes B \end{bmatrix} \right\| \\ &= \|P^2 + Q^2\|^{1/2} \|A^2 + B^2\|^{1/2} \leq \sqrt{2}, \end{aligned}$$

because $A^2 \preceq I$ and $B^2 \preceq I$. Hence the objective value is at most $\sqrt{2}$ at all levels.

For equality, take the Pauli matrices

$$Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Set

$$X_1 = Z, \quad X_2 = X, \quad Y_1 = \frac{Z + X}{2}, \quad Y_2 = \frac{Z - X}{2}.$$

Then $X \in \mathcal{C}_2(2)$. Also $Y_1 + Y_2 = Z$ and $Y_1 - Y_2 = X$, so $Y \in \mathcal{D}_2(2)$. The corresponding objective is

$$\frac{1}{2} (Z \otimes Z + Z \otimes X + X \otimes Z - X \otimes X),$$

the CHSH operator divided by 2, whose spectrum contains $\pm\sqrt{2}$. Thus the supremum is at least $\sqrt{2}$, completing the proof. $\square$

Consequences and Scope

By the source's Theorem 6.1 / Theorem 7.2 dictionary between these inclusion constants and measurement incompatibility, this also recovers the all-dimension complex value

$$s_{\mathbb{C}}(d, 2, (2, 2)) = \frac{1}{\sqrt{2}}$$

for two dichotomic measurements. The packet does not address $k \geq 2$ in Conjecture 5.2 / Conjecture 7.3, nor the separate four-dichotomic-qubit conjecture $s_{\mathbb{C}}(2, 4) = 2/\sqrt{13}$.

Novelty Check

The search on July 18, 2026 found no exact run duplicate for arXiv:2512.17706. This is a small subcase and is close to known matrix cube/matrix diamond dilation facts and the standard two-observable Tsirelson bound. Web searches for the source arXiv id, the label `kSimplexPlusLineOptimum`, matrix square inclusion constants with $\sqrt{2}$, matrix cube/matrix diamond phrases, and two-dichotomic compatibility-degree phrases found adjacent known literature but no explicit later resolution of the source conjecture. The claim here is therefore scoped only as a modest partial result.

References

- [1] A. Bluhm, E. Evert, I. Klep, V. Magron, and I. Nechita, *Inclusion constants for free spectrahedra with applications to quantum incompatibility*, arXiv:2512.17706.